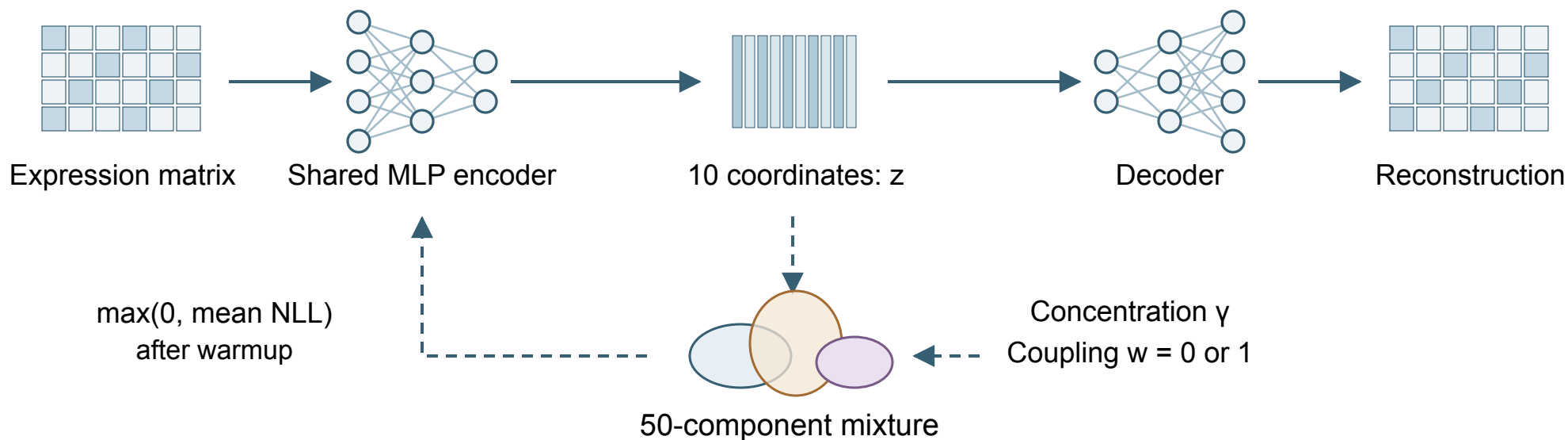
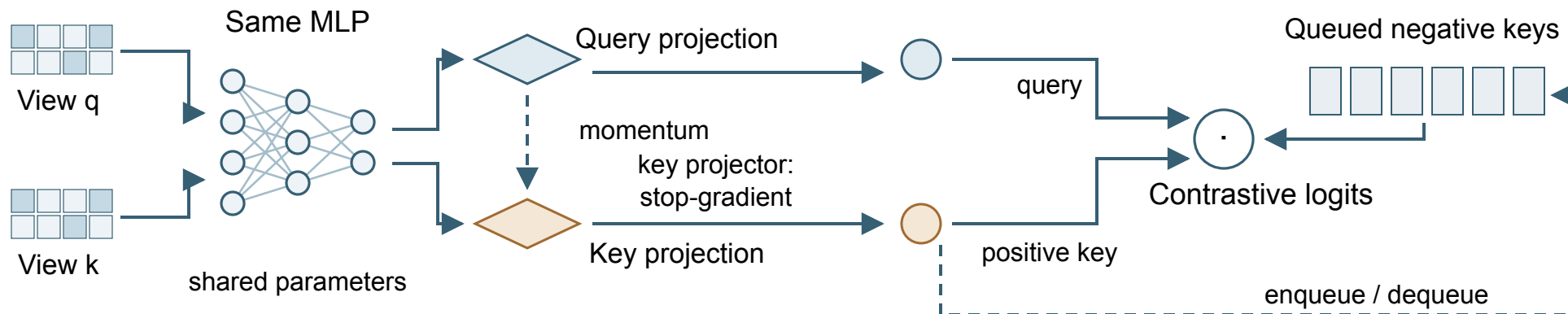


A

Reconstruction and mixture path

**B**

Stateful contrastive path retained in both coupling arms



Pairing at epoch 180 preserves queue, normalisation buffers, optimiser and random state.

Loss: $\text{MSE} + w \times \text{mixture} + \text{queue} + 0.5 \times \text{symmetric} + 0.3 \times \text{prototype}$